

Optimal DMO, Demand-Driven, Debt Structure: Time to Add SeLFIES and BEST Bonds?

By

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Abstract

Debt Management Organizations (DMOs) face interesting challenges with significant refinancing needed in the next few years (especially with maturing T-bills), and a decline in demand for long-term bonds from local defined benefit pension plans. This raises an interesting question: what is the optimal debt structure of DMOs? The traditional approach has focused on the cost-risk trade-off for DMOs, but we argue that this potentially myopic approach risks inheriting an unfunded future contingent liability. Ignoring the needs of goals-based investors, in this new paradigm of transferring investment risk, especially retirement risk (and other stochastic goals) to individuals could be a mistake and this paper quantifies that risk. Traditional bullet nominal or inflation-linked (IL) bonds were meant to repay principal. In reality, investors invest to achieve investment goals like retirement, saving for a child's education, health etc. and traditional bonds are risky for these goals. The successful creation and issuance of unique and innovative retirement and education bonds by Brazil in 2023, with very specific goal-matching cash flows (CFs), without a principal repayment, to good and sustained demand, alters the perspective of not only what bonds investors want, but also the optimal debt composition of DMOs. Using data from Brazil on a range of fixed rate, IL and CF-focused bonds allows us to put some values on the risks that DMOs might engender in the economy by not issuing more creative CF bonds and derive "optimal" demand functions. This new demand-driven optimal debt structure approach, also compliant with Islamic finance principles, might ensure that DMOs can meet varied domestic fiscal and financial goals.

Keywords and JEL Classification

Keywords: Stochastic Goals; Liability Driven Investing; Goals-Based Investing; Inflation-Linked Bonds; Optimal Debt Structure; DMO; Debt Management Organizations; Fixed Rate Bonds, VAT-Linked Bonds; Retirement Bonds; Education Funding Bonds; GDP-Linked Bonds; Sukuk Bonds

JEL Classification: G1, G2, G12, H63, H68

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1. I received no financial support from any entities
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3. There were no non-disclosure arrangements that have any relevance to this paper.

“Moreover, since the sun remains stationary, whatever appears as a motion of the sun is really due rather to the motion of the earth.”

“Those things which I am saying now may be obscure, yet they will be made clearer in their proper place.” Nicolaus Copernicus¹

Introduction:

The traditional challenges faced by debt management organizations (DMOs), in deciding on the optimal debt structure and annual debt issuance targets, have been exacerbated by widespread global quantitative easing (QE) being reversed, but also the closing of defined benefit (DB) pension plans, which threatens to affect demand for long-term bonds issued by DMOs.² Countries as diverse as the UK³, Japan⁴ and Italy⁵ are struggling to ensure long-term retail demand as institutional sources of funding are declining, forcing DMOs into issuing shorter maturity bonds, while carrying a meaningful Treasury bill (T-Bill) position. To summarize the challenges, OECD (2025) notes: “Sovereign bond issuance in OECD countries is projected to reach a record USD 17 trillion in 2025, up from USD 16 trillion in 2024 and USD 14 trillion in 2023..... Central banks continued to withdraw from debt markets through quantitative tightening in 2024. If current levels of debt are to be maintained, either existing investors will need to buy more debt or new, likely more price-sensitive, investors will need to enter the market, which could increase volatility.... Outstanding sovereign bond debt in OECD countries is projected to reach nearly USD 59 trillion in 2025, up from USD 55 trillion in 2024 and USD 54 trillion in 2023. Borrowing through treasury bills will comprise half of market borrowing in 2024-25, exceeding the ten-year average by 10 percentage points.”

This paper seeks to present a new framework for DMOs to consider the optimal debt structure as they may be missing a key retail and institutional demand segment, recently revealed by two bond innovations pioneered by Brazil in 2023, and that these innovations are a good deal for DMOs, as suppliers of these bonds, as well. These new instruments not only suggest untapped demand from domestic, local currency, long-term, retail and institutional investors,

but also offer the possibility of lowering longer-term macro-economic risks, that have not been considered in the typical objective functions of DMOs.

For example, the Dutch DSTA 2025 Outlook publication notes that the coalition government wants, “robust government finances providing both prosperity and stability”.⁶ It refers to the challenges the Netherlands faces from the move from the old pension system to the new WTP (“Future Pension Act”) system that moves from DB to defined contribution (DC). The UK DMO notes, “The DMO's remit is to carry out the Government's debt management policy of minimising financing costs over the long term, taking account of risk, and to minimise the cost of offsetting the Government's net cash flows over time, while operating in a risk appetite approved by Ministers in both cases.”⁷ The Italian MeF has issued BTP Italia, inflation-linked (IL) bonds targeted to retail investors, BTP Futura bonds targeted to retail investors (during Covid-19, with a step-up structure and a loyalty premium for long-term holders), and ideally fund long-term green infrastructure. MeF wants local buy-and-hold (BH) investors, as currently over 40% of its debt is held by foreigners. Finally, the Austrian OeBFA notes in its June 2025 Investor Presentation that the cornerstones of its funding goals are, “Diversity of funding sources (geographically and by instruments), including a dedicated retail product; Conservative debt management strategy; over 90% fixed rate; below 20% short-term debt; No foreign currency risk; and Well-balanced maturity profile.”⁸

A Cochrane (2022), quote motivates this paper: “[S]witching to a payoff perspective may change lots of procedures in finance.” The ideas presented here are particularly relevant and further grounded in reality as Brazil has issued two such cash-flow-focused bonds (CFFBs) in 2023 to good success – RendA+ (or “retirement income bond”)⁹ and Educa+ (“education bond”)¹⁰ based on Muralidhar (2015, 2016), respectively. Muralidhar and Merton (2020) term the retirement bonds, “SeLFIES” or Standard-of-Living-indexed, Forward-starting, Income-only Securities, and Muralidhar (2016) terms the education bonds, “BEST” or Bonds for Education and Student Tuition. What these CFFBs offer in their design, as articulated in Muralidhar (2016), is that instead of traditional bullet bonds focused entirely on principal repayment (and hence the term “bullet”), including in pricing these bonds in primary issuance

to be close to parity by selecting the then relevant coupon to match the equivalent yield for that maturity, CFFB pay a unit of a series of exact cash flows, tailored to the very specific and important investment goals of individuals (or in collective arrangements as in DC pension plans). For retirement, SeLFIES would offer \$10 real/year, payable monthly, for say 20 years (average life expectancy), starting at the retirement age, and ideally indexed to standard-of-living (Bodie, Merton and Samuelson 1992). BEST would offer \$10 real/year for say 4 years (5 in the case of Educa+), starting at age 18 (when a child goes to college), and indexed to tuition inflation.

Why does this make sense? Merton (2014) and Merton-Muralidhar (2020) has shown how T-Bills, Fixed Rate, Variable Rate and Inflation-Linked bonds are risky for these goals. For example, if an individual needed \$50,000 real/year in retirement or for a child's college education, they just needed to buy 5,000 of the respective bonds prior to the trigger event (with the flexibility to change their date etc.). In short, to achieve one's goal requires buying just multiples of a single relative risk-free security (Merton and Muralidhar 2020) as opposed to current practice of buying multiple securities to match goal cash flows and requiring costly and complicated financial engineering, and risky re-financing to satisfy the demand of long-term investors. For example, a 25-year-old saving for retirement must have a 60-year perspective (40 years of accumulation and 20 years of decumulation) and buying and managing current instruments for this cash flow stream is risky, costly and complicated, and beyond the comprehension of a relatively financially unsophisticated population. This is how CFFBs can greatly improve goal-achievement and risk management in a goals-based world. Merton-Muralidhar (2020) argue that these bonds are a good deal for governments too as (i) SeLFIES lowers the risk of retirement poverty (which will lead to bailouts and future contingent liabilities); (ii) provide matched funding for long-term (green) infrastructure; (iii) can be sold to dedicated, largely, BH domestic investors (reducing currency risk, market volatility, and risk of economic exposure to foreign actors); and (iv) extends the yield curve to ensure capital market development at maturities beyond where many DMOs, even in developed countries, can issue debt. If these claims by Merton-Muralidhar (2020) are true, they should feature in the optimal debt structure of DMOs, and we seek to prove this point in

the paper by quantifying the first point and proving the fourth point; the other two are obvious because they are in the goal statements of the DMOs mentioned above.

Our approach is similar to Guibaud et al (2015) who state that they offer a, “clientele-based model of the yield curve and optimal maturity structure of government debt.” They offer a review of the literature, which we do not repeat here for brevity, and state further that “[O]ur model differs because we allow for clienteles with different investment horizons.” But they miss the cash flow or relative wealth perspective to goals-based investing (GBI) that we cover in Section I. While they use the UK 2004 Pension Reform Act to motivate their paper, and discuss pension liabilities and how it affects duration choice, they ignore the fact that in a model where individuals and institutions care about relative wealth (i.e., stochastic wealth relative to stochastic liabilities), cash flows matter more so than “investment horizons”. Hence, the approach presented here is very different from previous research. They also use an unobserved risk aversion parameter, whereas the demand-for-assets model we present below is based on Muralidhar (2000), and reflects how institutional and retail investors actually behave. The realistic aspects of this approach are discussed in detail in the preferred habitat general equilibrium model of term structure/asset prices, asset allocation and wages of Muralidhar (2025a and 2025b).

The paper is structured as follows: Section I provides a quick overview of investment reality in a GBI world to demonstrate that effective goal management by individual (and institutional investors) is about matching cash flows and NOT preserving wealth (as assumed in the starting assumption of finance theory that “investors maximize the expected utility of wealth). Section II provides details and data on Brazil’s RendA+ and Educa+ CFFBs, along with Brazilian fixed rate and IL bonds, which will be critical to estimating risks engendered by not issuing CFFBs (in other countries). Section III then provides some basic formulae to demonstrate the reason for altering the recommendations for “optimal debt structure”, and Section IV uses Brazil live data to demonstrate the results. Section V considers some alternative design aspects for such bonds, including indexation to a Value-Added-Tax (VAT) index and why it may benefit countries and provides some discussion of the results. Section VI concludes.

I. The Reality of Investing in a GBI World

Financial innovation is often based on the academic literature and one of the major challenges in finance is that the seminal papers (Markowitz 1952, Tobin 1958, Sharpe 1964) were based on assuming that “investors maximize the expected utility of wealth”. However, Sharpe-Tint (1990), Merton (1993, 2014) change the perspective to relative wealth (or funded status), where the investor cares about NOT wealth, but the cash flows that pertain to the goal. An alternative way to understand this point is that wealth for wealth’s sake has no meaning; all our goals have specific cash flows (in aggregate called “liabilities”), and hence investments must be tailored to match these cash flows or liabilities. Interestingly, even Modigliani-Brumberg (1954), where savings are needed to finance one’s retirement cash flow when labor income ceases, is probably the earliest representation of GBI. Muralidhar (2016, 2023) changes the paradigm to focus on satisfying the cash flows needed for a range of goals (as proposed in Merton 2007) as opposed to wealth, as that is what is truly relevant for every investment portfolio. In a GBI approach, one must start with the stochastic cash flows associated with the goal (as a way to benchmark that goal on a daily basis as demonstrated in Muralidhar-van Stuijvenberg 2005 in the context of DB pensions). If such a benchmark, with a clearly observed price exists, then one can calculate the risk-return (and correlation) characteristics of the stochastic goal. Historically, such benchmarks for GBI have been hard to construct for retail investors.

Following Merton (2014), Muralidhar (2016) highlights that for most (retail) goals and some institutional goals, these goal cash-flow replicating assets (or portfolio of assets) do not exist, and hence argues for the creation of such instruments. In order to create such instruments, one must find a natural issuer for these instruments so that supply meets demand. Muralidhar (2016) shows how a natural issuer for the retirement goal would be a government, and a natural issuer for education could be private institutions or governments, and so on for other goals. For example, the natural issuer for the absolute risk-free asset would also be governments, and Muralidhar (2018) extends this idea of CFFBs further to show how cohort-level longevity risk can be hedged in retirement accounts with a natural issuer (governments).

SeLFIES (or BEST) bond is a single, simple, liquid, low-cost, relatively low-risk (government-issued) instrument, easy-to-understand for even the most financially unsophisticated individual, because it matches the desired real retirement income profile of individuals and embeds accumulation, decumulation, compounding and inflation-adjustments.

Other research has highlighted how corporate entities could issue financial securities to hedge other goal cash-flow risk (like hedging health cost risks, which have a more equity component to them). The existence of entities to issue such debt or equity securities addresses the critical assumption in this paper that CFFBs not only have demand from investors, but have a credible supplier with an offsetting hedge. Brazil's Tesouro Nacional successfully issuing the RendA+ retirement bond and Educa+ education bond, with significant and growing demand (over 500,000 investors voluntarily buying each of the securities in less than 2 years), not only proves this point, but the fact that other countries are considering these innovations¹¹ shows that these are global issues, and not just isolated to Brazil. In short, there is clearly non-zero demand for CFFBs.

II. Brazil's Bond Innovations and a Different Preferred Habitat Perspective

Brazil issued two innovative bond instruments in 2023 – RendA+¹² and Educa+¹³, based on Muralidhar, Ohashi and Shin (2016) and Muralidhar (2016), respectively. This makes sense as life-cycle models in economics, as initially shown in Modigliani and Brumberg (1954), and much more forcefully in Merton (2014), is all about managing cash flows, and not wealth, and in turn establishes the “preferred habitats” (Muralidhar 2025b). The cash flows of generic Brazilian retirement and education bonds are provided in Figure 1, but there is no requirement that these CFFBs be solely debt instruments. We focus on the debt instruments as we are trying to ascertain if this demand can/should be satisfied by DMOs.

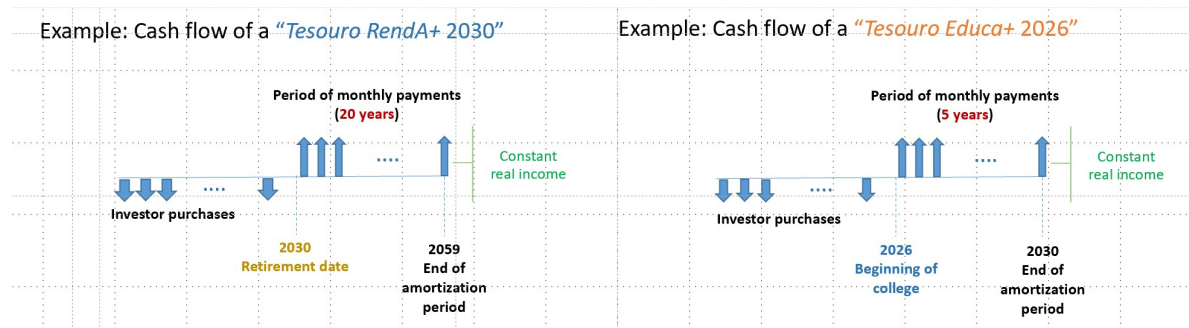


Figure 1: Cash flows of retirement (20 years starting in 2030), and education (5 years starting in 2026) bonds issued in Brazil (Source: *Tesouro National presentation to International Retail Debt Management Symposium, Washington DC, November 2024*).

Each of these instruments was tailored to the unique nature of these cash flows in Brazil, and both were IL to IPCA+, as IPCA+ is the dominant inflation index in Brazil (we discuss alternative indexation in Section IV). Brazil launched Renda+ on their Treasury Direct platform, allowing individuals to purchase these bonds off either the internet or an app on their phone, much like Austria, and in slices as small as €1/time (fractional bonds). They issued 8 different Renda+ series – with cash flows starting in 2030 (for a 60-year-old today), 2035, ..., 2065 (for a 25-year-old). Purchases are voluntary, but because prices are quoted three times a day, they are liquid and individuals can reverse or change their date of retirement or target supplemental income at will. They are also inheritable and just require answers to the two questions: target retirement date and desired real cash flow in retirement. Moreover, the Brazilian Treasury went further and allowed individuals to purchase these bonds via gift cards, so benevolent employers (individuals) can gift employees (friends) guaranteed real retirement income. As of May 2025, 312,450 individuals bought these bonds for a value exceeding R\$5 billion. This innovation received the runner-up award for innovation at the Global Pension Summit in the Netherlands in 2023. In addition, this innovation has led to a complementary innovation, Educa+, an education financing bond, with over 150,000 investors and raising approximately R\$1.4 bn. Additionally, Brazilian Treasury offers detailed information about the number of investors and bonds purchased for each bond series (PH). These innovations are being considered in countries as diverse as US, UK, India, Turkey, etc.

Table 1 provides historical data on RendA 2060 and 2035, Educa 2030 and Tesouro Prefixado 2031 (fixed rate) and IPCA+ (IL) 2035 and 2045. We include all available daily data from January 1, 2023-October 29, 2024, except for Educa 2030, which was only launched on August 1, 2023. We provide (arithmetic) annualized returns, volatilities, durations, and current yields (where available). We also provide a correlation matrix of returns, which will be critical for the discussion/quantification of optimal debt structure and government contingent liability risk in Section III.

Volatilities							
	Educa 2030	Educa 2040	RendA 2035	RendA 2060	Tesouro IPCA+ 2035	Tesouro IPCA+ 2045	Tesouro Prefixado 2031
Ann Return	0.25%	-6.64%	-7.41%	-21.67%	-2.55%	-9.40%	-5.49%
Ann Risk	7.13%	14.45%	14.88%	35.98%	9.60%	16.60%	10.23%
Duration	7.84	12.81	17.27	42.14	9.18	18.5	4.8
Yield as of June 2025	7.3	7.02	6.93	6.86			
Correlations							
	Educa 2030	Educa 2040	RendA 2035	RendA 2060	Tesouro IPCA+ 2035	Tesouro IPCA+ 2045	Tesouro Prefixado 2031
Educa 2030	1.000	0.938	0.936	0.873	0.963	0.903	0.759
Educa 2040		1.000	0.988	0.923	0.952	0.978	0.703
RendA 2035			1.000	0.934	0.956	0.978	0.709
RendA 2060				1.000	0.888	0.914	0.660
Tesouro IPCA+ 2035					1.000	0.925	0.730
Tesouro IPCA+ 2045							0.674
Tesouro Prefixado 2031							1.000

Table 1: Historical Data on Key Bonds Considered for this Analysis (Source: Tesouro Nacional and Natalia Cysne).

Figures 2A and 2B plot the duration and real yields of the current RendA and Educa issues (based on start dates of cash flow payments). In both charts, we notice relatively high real yields, but also a downward sloping term structure as we go out further on the duration spectrum. Figure 3 combines the two data series and plots duration versus real yield of both instruments to show how this covers the term structure from 3 to 47-year duration. This has interesting implications for Brazil (and other DMOs), if lowering the interest cost and diversifying sources of funding is the goal.

Optimal Debt Structure

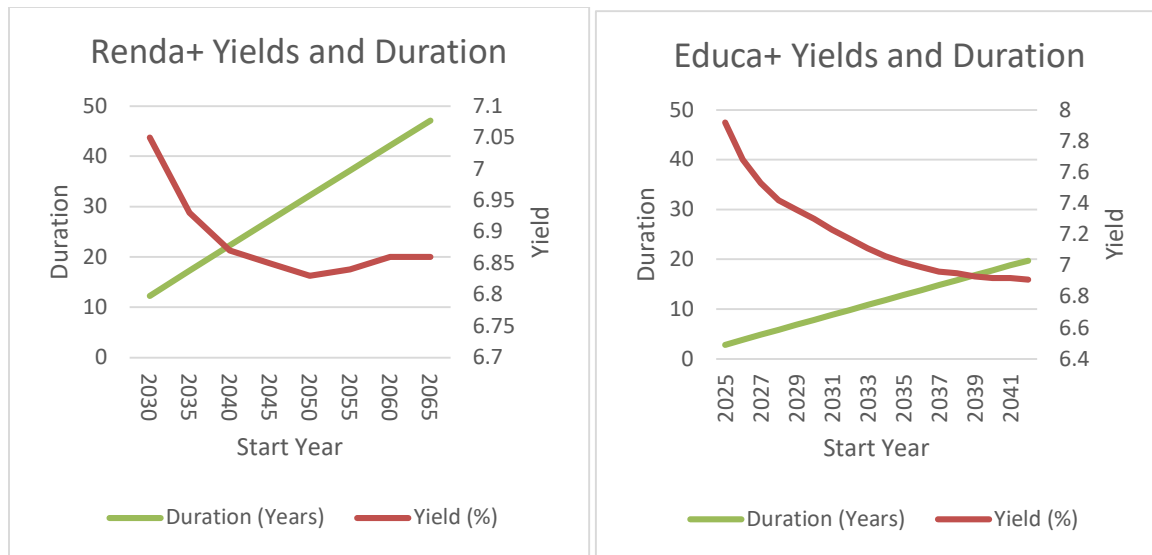


Figure 2A and 2B: Duration and Real Yields of RendA+ and Educa+, based on CF Start Years (Source: Tesouro Nacional).

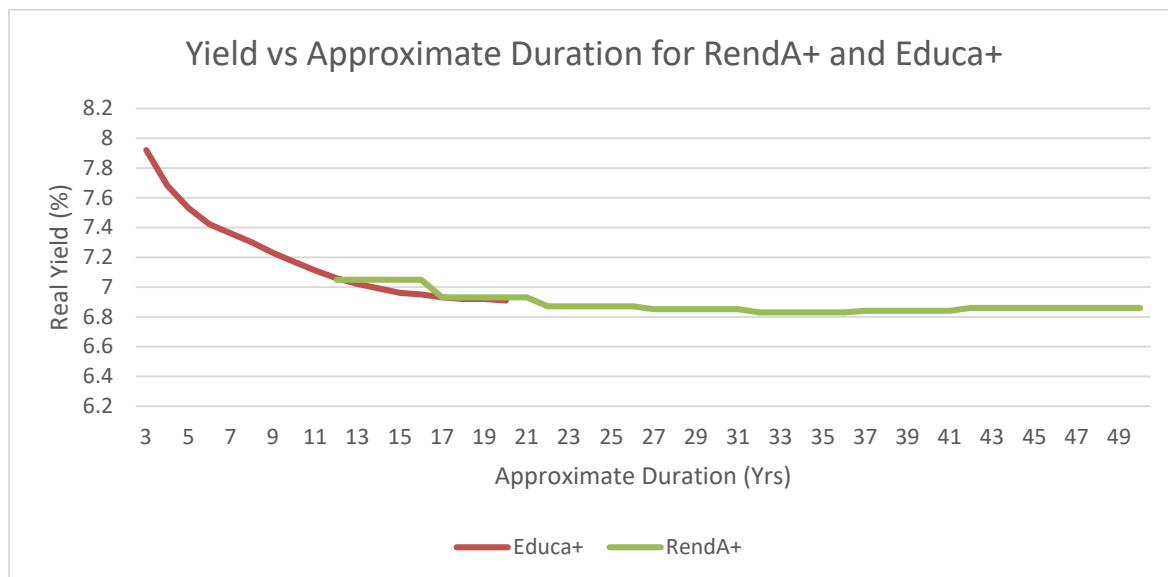


Figure 3: Duration vs Real Yields of RendA+ and Educa+, as of June 2025 (Source: Tesouro Nacional).

III. Optimal Debt Structure

i. Background

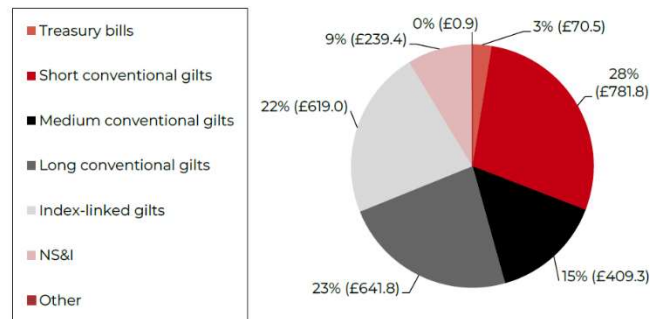
OECD (2025) reports that “the fixed-rate bond share of the outstanding debt remained largely stable at 76% in 2024 with T-bills’ share remaining at nearly 15% (Figure 1.7, Panel A), ... while the share of inflation-linked bonds will continue a gradual decline, approaching 7% in 2024.” Obviously, each country will have a different debt structure depending on its own goals/objectives, and market access/willingness to fund debt. For example, as World Bank (2022) notes, countries with higher inflation have a harder time issuing long-term, fixed rate debt, and issue more IL-debt for obvious reasons.

As we noted in Section II, Brazil has a greater ability to extend maturities further down the yield curve with IL bonds than fixed rate bonds (and hence our choice of Prefixado 2031 and IPCA 2045 in the Table 1). World Bank (2022) coincidentally provides the Brazilian Debt Management Strategy as shown in Table 2 below, and this will be useful as Brazilian data on CFFB and “old” bonds was provided in Table 1. What is interesting from Table 1 is the level of detail for each type of instrument (and Table 2 was the strategy prior to the 2023 creation of CFFBs, and hence their absence in the table). Most fascinating is that the “average maturity” was just 5.5 years in 2022, but as Table 1 shows, Renda 2035 and 2065 have durations ranging from 17+ years to 40+ years, respectively, giving Brazil the opportunity to extend duration should it choose to do so with a downward sloping real yield curve (Figure 3). In other words, these instruments allow DMOs to extend the duration of their borrowing should they seek to do so, and if there is sufficient demand (which we demonstrate/try to estimate in Section III.iii and Section III.iv below). Strictly for comparison purposes, we present the outstanding gilt debt for the UK in Table 3, which has, at end-December 2024, the average maturity of the total stock of 14.4 years. Clearly, the debt composition of the UK DMO is different from that of Brazil, especially since the UK DMO has benefitted from local DB pensions hedging liabilities with long-term debt. The DB pension market is less mature in Brazil.

TABLE 7.1. Brazil: Debt Management Strategy

	Long-term targets	
	Reference	Range
Composition (%)		
Fixed rate	40	+/- 2
Inflation linked	35	+/- 2
Floating rate	20	+/- 2
FX	5	+/- 2
Maturity Structure		
% maturing in 12 months	20	+/- 2
Average maturity (years)	5.5	+/- 0.5

Source: Data from the Ministry of Finance, Brazil.

Table 2: Brazilian Debt Structure (Source: World Bank 2022).**Chart A.1 Composition of central government sterling debt in % and £ billion (end-December 2024)^{1,2}**

¹Figures may not sum due to rounding. Nominal uplifted values.

Table 3: UK Debt Structure as of December 2024 (Source: HM Treasury 2025).

ii. Analytical Framework

Consider the following definitions to make the discussion easier. Define B_i to be the version of the bond that can be issued,

where $i = N$ for nominal bullet (or fixed rate bond), IL for Inflation-Linked, R for retirement (RendA+), E for education (Educa), G for GDP-linked (Shiller 1993, 2005 discussed in Section IV), T for T-Bill, W for Wage-Indexed (because Uruguay issued these bonds), C for Consols (Cochrane 2015 discussed in Section IV), and V for VAT-Indexed (discussed in Section IV). For simplicity, assume that B (total debt outstanding) $= \sum_i B_i$, and we could assume that B_O (or “Old” Bonds) $= B_N + B_I + B_T + B_G$, and B_{CF} (or CFFBs) $= B_R + B_E$. Assume ω_i^D is the demand (in percentage terms) of the i th bond (see preceding list for various types, and for example the Nominal Bond weight would be ω_N^D , and such that $\sum_j \omega_j^D = 1$). Assume r_j is the annualized real interest and ω_j^S is the supply of the j th bond. The total annual interest cost $r_B = \sum_j r_j * \omega_j^S$.

It appears that the traditional DMO objective is to

Minimize r_B subject to B being sufficiently diversified (1)

And with possibly of keeping the volatility of r ($\sigma(r)$) low, and also targeting possibly a duration/maturity profile, as is the case in the Brazilian example in Table 2. As HM Treasury (2025) highlights, the goal is to minimize 5 risks: interest rate risk – interest rate exposure arising when new debt is issued; refinancing risk; inflation risk; liquidity risk and execution risk.

However, the one issue not incorporated into these objectives is the risk of issuing securities to institutions and retail investors that engender a duration/cash flow mismatch relative to their goals. This is important, because as Modigliani-Muralidhar (2004) and Merton-Muralidhar (2020) note in the context of retirement that if individuals retire poor (below some basic income threshold) because they either did not save enough or invested in securities that are risky relative to the cash flow of their retirement goal, then the governments will have to bail them out. This is a contingent liability that will add to the DMO financing needs in the future, and needs to be incorporated into the objective function. This triggering of a contingent liability and an unexpected bailout was witnessed in Chile and other countries in the late 2010s. Low interest rates led to very low retirement income from accumulated balances in private pensions, and coupled with covid-19 challenges, led to increased projected fiscal expenditures

(Inzunza, A. and C. Madeira. 2025). Merton-Muralidhar (2015) attribute this to a flawed monetary policy that focused on absolute wealth and ignored the pernicious impact on relative wealth of low interest rates. The same could be argued for financing a child's education as witnessed by the high levels of student debt in America, that some Administrations have tried to write off, but that just increased the borrowing needs of the DMO, and hence are a contingent liability that we attempt to quantify in Section III.iv.

We restate the objective of the DMO (as supplier of bonds) as:

Minimize r_B , AND future fiscal costs from investors not achieving social goals,
subject to B being sufficiently diversified and managing the other five risks (2)

The future fiscal costs of investors not achieving their retirement goals or not being able to save enough for a child's education will be a function of the tracking error between the securities issued by the DMO (and their optimal portfolio structure for given risk-adjusted return goals), and the cash flows they need for the goals, where tracking error is defined as

$$TE(A, L) = \sqrt{[\sigma^2(A) - 2 * \rho(A, L) * \sigma(A) * \sigma(L) + \sigma^2(L)]} \quad (3)$$

Where A = asset portfolio, L = the stochastic goal, $\sigma(*)$ is the volatility, and $\rho(A, L)$ is the correlation between the two. Tracking error is relevant because even if the asset portfolio has the same expected return as the stochastic goal, the lack of correlation and differences in volatilities imply that even if an investor saves exactly what they need to, $TE(*)$ measures a one-standard deviation event from the target goal of retirement or education of one's child. For example, there are many silly advertisements/research papers about how in 2025, the target wealth one needs for retirement in the United States is \$1.26 million¹⁴ – a one-standard deviation loss for a single individual in an economy would be, using this silly wealth metric, \$126,000 - \$300,000 (assuming tracking errors of 10% and 25%, respectively). There are an estimated 55 million retirees in the US. Hence, high tracking error in retirement portfolios can translate very quickly into multi-billion-dollar contingent liabilities for DMOs. In short, TE provides an estimate of the shortfall risk for each of the goals (retirement and education), and hence the cost of the contingent liability of the government.

Examining the demand side of the equation, Muralidhar (2000), following Modigliani-Modigliani (1997) and Sharpe-Tint (1990), solves for the optimal portfolio assuming that investors have the following risk-adjusted return objective (Equation 4), which appears to be case from observing the Investment Policy Statements (IPS) of institutional investors and surveys of retail investors (Muralidhar 2025). Assuming A is the composite asset portfolio made up of multiple instruments, and L is the goal replicating asset, replicate by either a portfolio of assets as in the case of a DB pension or a single asset using SeLFIES or BEST bonds, rational goals-based investors, worried about the skillfulness of agents/asset managers/advisors they have hired to manage their portfolios, will specify their objective function as equation (4) subject to the constraints in equations (5), (6) and (7):

$$\max E[r(A) - r(L)] \text{ or equation (1)} \quad (4)$$

subject to

$$E[r(A)] = a_{\tau/L}^P \times E[r(P)] + l_{\tau/L}^L \times E[r(L)] + (1 - a_{\tau/L}^P - l_{\tau/L}^L) \times r(F) \quad (5)$$

$$\sigma(A) = \sigma(L) \quad (6)$$

$$\text{and TE}(A, L) = \text{TE}(\text{Target}, L) = \tau \quad (7)$$

where P is a risky portfolio (e.g., a fixed rate bond with a duration of 5 for say a pension fund with a target duration of liabilities of 15 or an equity index), and F is the traditional absolute risk-free asset of finance theory with zero volatility and correlation to other assets. Solving for $a_{\tau/L}^P$, $l_{\tau/L}^L$ and $(1 - a_{\tau/L}^P - l_{\tau/L}^L)$ is effectively solving for ω_j^D . For simplicity, we will assume that F is a proxy for T-Bills, as we do not have a continuous time series of T-Bill prices in Table 1. The solution to this optimization is provided in Appendix II and a typical Tracking Error (Target) = 2%-3% annualized, reflecting the low-risk investors typically want to take relative to their Strategic Asset Allocation (SAA). Investments are plagued by principal-agent problems (Allen 2001) and the IPS of investors give us values in the 1-3% range (Muralidhar 2023 and 2025a).

iii. Tracking Error Calculations – with Old Bonds only

We take the case of Brazil and assume that there are two types of investors – Retirement and Education investors, whose target cash flows are represented by B_R and B_E , respectively. Moreover, we assume that there are two retirement investors: a 55-year-old who will retire in 2035 (with cash flows represented by $RendA+ 2035$, with duration of 17 years), and a 30-year-old that will retire in 2060 (with cash flows represented by $RendA+ 2060$, with duration of 42 years). Moreover, there is one education investor, with cash flows best represented by $Educa+ 2030$ (with a duration of 4.8).

In the first scenario, we assume that the only instrument issued is the Brazilian $IPCA+ 2045$ (with a duration of 18.5), since we have assumed that the cash flows of the 3 investors are IL. As a result, in this case, 100% of the assets are invested in $IPCA+ 2045$, and Table 4 demonstrates the implied tracking error (and hence future risk to the government/DMO from a mismatch of durations/cash flows between what investors need and what the DMO has issued). The duration of this bond is not too different from the guesstimated duration of the UK DMOs outstanding debt structure in Table 3. In short, with the exception of the retiree in 2035, with a duration close to that of $IPCA 2045$, all investors will be taking excessive risk relative to their goals, which increases the likelihood of a future bailout. Even for the 2035 retiree, the annualized tracking error is higher than a typical annualized target tracking error of most retirees/institutions.

TE (I2045 vs R2060)	21.87%
TE (I2045 vs E2030)	10.62%
TE (I2045 vs R2035)	3.74%

Table 4: Annualized Tracking Error Caused by Investing in $IPCA 2045$ for 3 Goals.

For completeness, Table 5 provides the annualized tracking error of other variations, including the annualized tracking error from holding T-Bills, which is a non-trivial component of overall DMO debt outstanding, and the risk of investing in a CF bond that is different from the CFs of one's goal. The top part of Table 5, validates a point made in Merton (2014) and Merton-Muralidhar (2020), that the “absolute risk-free” asset of finance theory, or F , that ostensibly preserves wealth, is very risky from the perspective of stochastic goals. This was the basis for arguing for creating these instruments in the first place (Muralidhar 2015, 2016).

TE (F vs R2060)	35.98%
TE (F vs E2030)	7.13%
TE (F vs R2035)	14.88%
TE (E2030 vs R2060)	29.70%
TE (E2030 vs R2035)	8.59%

Table 5: Annualized Tracking Error from Investing in “Risky” Securities for 3 Goals.

Finally, Table 6 provides the annualized tracking error assuming 100% of the portfolio is held in Prefixado 2031 (with a duration close to 5, which is near Brazil’s target duration in Table 2). As can be expected, given the duration mismatch, the lowest annualized tracking error is for the 2030 education investor, and all others suggest massive contingent liabilities.

TE (Prefixado vs R2060)	30.22%
TE (Prefixado vs E2030)	6.69%
TE (Prefixado vs R2035)	10.50%

Table 6: Annualized Tracking Error from Investing in Prefixado 2031 for 3 Goals.

In short, with live data from Brazil, we are able to prove our first key point; namely, that with the current issuance of bullet nominal, IL and T-Bills, DMOs have increased the risks of future bailouts and will need to consider the innovations pioneered by Brazil for a more optimal debt structure if they believe GBI is the future. Even though there may be multiple instruments on offer for diverse goals-based investors, the complexity of financial engineering, and the resulting cash flow mismatch from bonds that focus on repayment of principal for goals that are strictly income-only (IOs), especially among financially unsophisticated citizens, could lead to trouble in the future.

iv. Tracking Error Estimates – with CFFBs and Old Bonds

Since Brazil issued RendA+ and Educa+ (and a series of them), we can use the optimal portfolio allocations for these 3 investors to investigate what would be the demand, given say an annualized target tracking error of 3% for all three goals, and the resulting “optimal” debt structure that is demand-focused, and the resulting duration profile of outstanding debt.

We start first with assuming that IPCA+ 2045 is now the risky asset (P in equation 5) and solve for each of the 3 goals and aggregate the results in Table 7. Relative to a naïve average target duration of 22.42, because of the desire to take on risk (via the annualized target tracking error of 3%), the target duration of optimal demand is 22.88. If DMOs would seek to match issuance to correspond to demand, there is a case for T-Bills to be approximately 10% of total issuance, IPCA+ 2045 to be approximately 57%, and the balance largely dominated by young retirement investors (despite the fact that we assumed that the three investors had equal wealth, which is not always the case as older investors have larger accumulated balances).

Asset	Total Sales	Optimal Debt Structure	Duration	Weighted Duration
F	28.56%	9.52%	0.00	0.00
Tesouro IPCA+ 2045	171.21%	57.07%	18.50	10.56
RENTA 2060	80.88%	26.96%	42.14	11.36
EDUCA 2030	4.78%	1.59%	7.84	0.12
RENTA 2035	14.57%	4.86%	17.27	0.84
Total	300.00%	100.00%		22.88

Table 7: Optimal Debt Structure with IPCA 2045, T-Bills and 3 Goal-Replicating Bonds.

Table 8 conducts the same analysis using Prefixado 2031, but given the shorter duration and higher annualized tracking error relative to the goals (Table 6 vs Table 4), not only is there a lesser need for Old bonds, but also investors have to leverage (as evidenced by the aggregate short position in F) their portfolios to achieve their goals. This leverage of the financial system in 2025 is evident in the multi-trillion-dollar balance sheets of central banks. Moreover, the target duration based on optimal demand is lower than the target duration averaged across the three goals. The net effect is that weight of $B_O = 30\%$ and $B_{CF} = 70\%$, with 30% in Renda+ 2060, approximately 25% in Renda 2035, and approximately 15% in Educa+ 2030.

Asset	Total Sales	Optimal Debt Structure	Duration	Weighted Duration
F	-33.01%	-11.00%	0.00	0.00
Tesouro IPCA+ 2035	124.34%	41.45%	4.80	1.99
RENTA 2060	92.34%	30.78%	42.14	12.97
EDUCA 2030	43.18%	14.39%	7.84	1.13
RENTA 2035	73.15%	24.38%	17.27	4.21
Total	300.00%	100.00%		20.30

Table 8: Optimal Debt Structure with Prefixado 2031, T-Bills and 3 Goal-Replicating Bonds.

In short, what these two simple examples were meant to show, is that issuing Old debt in a GBI world could lead to CF (and duration) mismatches, but also a missed opportunity by DMOs to extend duration and build a loyal, BH, domestic/local currency, retail investor base, that can be serviced using innovative new technologies that deliver such instruments electronically and via phones to ensure financial inclusion (as they can be bought in fractional shares of as small as \$1/time as is the case in Brazil).

IV. Extensions and Discussion

Before considering additional detailed issues about structuring these bonds, indexation, accounting treatment etc., we want to acknowledge that these CFFBs are also “Sukuk”, thereby opening up additional demand from investors with an Islamic finance perspective (Muralidhar, Gerz and Derici 2024). This was not incorporated as part of the demand function in equation (5), but transforms the optimization completely because for investors in Sukuk or shariah-compliant portfolios, Old bonds would not be acceptable. Hence the approach presented here is applicable to DMOs in Islamic finance countries as well.

i. Alternative Debt Structures Tied to Wages, GDP and Per-capita Consumption

There has been a history of (academic) recommendations for creating new financial instruments, not always ending up with a successful launch, in part because they were based on academic theory or because they were designed from the perspective of the issuer, but some also from the perspective of the investor, and even both.

In the last category of satisfying both and probably one of the first ideas, Merton (1983) had argued for the creation of a unique pension bond that should be issued by the government, indexed to standard-of-living (e.g., per capita consumption), and sold to the Social Security Administration, so that social security pensions could be indexed to standard-of-living and not just inflation. The benefit to investors is that the critical risk of retirement – preserving real standard-of-living – is hedged, and the government is hedged too, as this bond is tied to potential tax receipts. This innovative idea has been modified in the design of SeLFIES (Muralidhar and Merton 2020), and in part, the model on which Brazil's Renda+ was launched. Brazil chose to initially launch an IPCA+-linked bond given its relative success and long track record), but could consider alternative indexation alternatives if Renda+ continues to be successful. The indexation of Merton (1983) was critical, not the relatively unspecified cash flow, and that was recast and more clearly specified in Muralidhar, Ohashi and Shin (2016). In discussions with some countries, it is clear that linking to per-capita consumption is much too complicated, so a VAT-linked bond has been considered. In such a case, DMOs are effectively adding one more component to the objective function in equation (2); namely, effective cash flow matching between receipts (VAT) and payments on B_V .

In the first category of favoring the issuer, Shiller (1993, 2005) advocated for the creation of GDP-linked bonds, which while useful for the issuer (supplier), because it allowed for lower debt service during periods of economic stress, had no obvious demand for the cash flows from institutional or retail investors (who are already long GDP growth in their equity allocations or their individual lives). These bonds were largely unsuccessful despite being issued by a few European countries. In effect, by focusing first on the demand side of such cash flows in this paper (as opposed to the supplier side as in Shiller 1993), there is a slightly higher likelihood of success as evidenced in Brazil.

In the both category, Uruguay issued wage-indexed bonds targeted to institutional investors because the pension law requires pensions to be indexed to wages. Absent such instruments, there would have been no hedge for insurance companies to offer such pension products, and hence Uruguay DMO issued multiple maturities at the shorter-end of the yield curve,

reflecting such demand.¹⁵ Uruguay is hedged in that income taxes are levied on wages, but this linkage does not protect investors from real standard-of-living risk.

Finally, Muralidhar (2016) argued that BEST should be indexed to tuition inflation as tuition inflation has far exceeded consumer price inflation. This indexation is feasible if these bonds are issued by private or public universities or colleges as their tuition receipts reflect tuition inflation and hence issuing such bonds is an effective way for US universities to fund themselves. This instrument has become particularly relevant in 2025, as the Trump Administration has withheld large grants, foreign student attendance is probably going to decline and endowments are illiquid because of investments in private assets (Muralidhar 2025c). Brazil issued Educa+ linked solely to IPCA+ for much the same reason as the indexation of RendA+, but clearly if the true cash flow for the education goal incorporates tuition inflation in the US and UK, the estimates of annualized tracking error in Section III.iii and Section III.iv, are understated.

ii. Thoughts on Inflation-indexation in the Developed World

While IL bonds have been successful in the United Kingdom, because an attempt is made to pay real pensions from UK DB plans, it has been less successful in the United States¹⁶, even discontinued in Canada¹⁷, and not even considered in the Netherlands¹⁸ – all large pension markets. Boermans and Swinkels (2025) offer a global perspective on demand for IL Bonds. There are a number of challenges that DMOs face with these bonds, and that the UK will inherit, as pension arrangements move from DB to DC. Hence, there is a need to innovate and countries like the Netherlands and Canada that view IL bonds as “expensive” are myopic because not issuing such bonds transfers real standard-of-living risks to individuals without giving them the financial instruments to hedge such risks. That mismatch between what the demand side of the CFFB market needs and what DMOs offer, will only raise the contingent liability.

First, there is a complete cash flow mismatch between the payment schedule of TIPs, with a concentrated principal repayment and what investors need in retirement (an IO profile), as

noted in Merton-Muralidhar (2020). Additionally, these bonds pay coupons per traditional payment schedules of quarterly or semi-annually, whereas investors need cash flows monthly to pay bills in retirement. Brazil captured this aspect in the design of RendA+ and Educa+, and have an additional feature; the retirement bond pays directly into their bank accounts on the same day that they receive their social security benefits monthly, thereby allowing this bond to be viewed as a complement to social security, and less of a financial instrument. Finally, in Brazil, the government offers the liquidity if individuals seek redemptions, whereas the bid-ask spread in the free market could be quite wide for less liquid TIPs.

iii. The weird case of the high debt EU countries

Interestingly, while the case for issuing SeLFIES has been presented to Treasury/government officials in France (Merton, Muralidhar, Martellini 2018), Italy (Fano, Galli, Muralidhar 2023), Spain (Herce, Merton, Muralidhar 2018), etc., countries with high debt-to-GDP ratios in the EU, governed by the EU borrowing rule, are averse to issue longer maturity SeLFIES debt because of a very naive accounting rule. According to these officials, say BTPPI 2060 (the Italian SeLFIES version), which would pay €5 real/year for 20 years and targeted to a 30-year-old, would engender €100 real of projected payments, which would be the accounting treatment.

Given an upward sloping real yield curve, DMOs in these countries would only receive say €50 from the sale of these bonds today. In short, they will receive less funds (€50) to meet their outstanding debt payments (€100) and budget deficits, despite the fact that these bonds achieve many of their stated objectives in the 2024-2025 Business Plans. They would much rather issue a bullet par bond, knowing fully well that SeLFIES, RendA+, BTPPI, etc., are useful instruments. Many of these countries are also part of the Retail Issuers consortium and have heard directly from Brazil's Tesouro Nacional officials on the success of RendA+ and Educa+. In other words, such bonds might be suited for issuance by countries like Austria, Estonia etc., with lower debt-to-GDP ratios, because there is a first-mover advantage, and any European citizen can buy these bonds and incur no currency risk. In a recent Op-Ed, Muralidhar and Ferreira (2025) argue that Portugal should consider issues SeLFIES/RendA+ because they are PEPPs (and the former finance minister of Portugal is the EU Commission

for Savings and Investments and seeks to create PEPPs). In this case, a naive accounting rule is leading to sub-optimal debt structure of DMOs.

iv. Extensions

The simplistic example provided in Section III needs to be complemented with a more comprehensive analysis of not only what critical goals do investors have (i.e., we assumed just two, but fortunately the Brazilian data offers the chance to examine 8 different retiree cohorts and 18 education cohorts), but also a quantification of the potential demand in currency terms. We have not incorporated the number of individuals in each of these cohorts and more importantly, the level of wealth accumulation by each cohort. So, while there may be a lot of young people in developing economies, their level of wealth accumulation is low thereby altering the optimal debt structure that a DMO might seek to target. In aging societies like the US, Japan, Korea, Italy, etc., with a sizable population reaching retirement with substantial savings, the lack of long duration, CFFBs should be a source of concern for DMOs and Treasury officials and an opportunity to innovate.

V. Conclusion

GBI is slowly becoming the norm for investors, and investors have multiple stochastic goals. DMOs face interesting challenges with significant refinancing needed in the next few years (especially with maturing T-bills), and a decline in demand for long-term bonds from local DB pension plans. This raises an interesting question: what is the optimal debt structure of DMOs? The traditional approach has focused on the cost-risk trade-off for DMOs, but this paper has shown that the traditional approach is potentially myopic and risks inheriting an unfunded future contingent liability and missing unique opportunities to issue long-term CFFBs. Ignoring the needs of goals-based investors, in this new paradigm of transferring investment risk, especially retirement risk (and other stochastic goals) to individuals could be a mistake and this paper quantifies that risk. Traditional bullet nominal or IL bonds were meant to repay principal. In reality, investors invest to achieve investment goals like retirement, saving for a child's education, health etc. and traditional bonds are risky for these goals.

The successful creation and issuance of unique and innovative retirement and education bonds by Brazil in 2023, with very specific goal-matching cash flows (CFs), without a principal repayment, to

good and sustained demand, alters the perspective of not only what bonds investors want, but also the optimal debt composition of DMOs. Using data from Brazil on a range of fixed rate, IL and CF-focused bonds allowed us to put some values on the risks that DMOs might engender in the economy by not issuing more creative CF bonds, and derive “optimal” demand functions. This new demand-driven optimal debt structure approach, also compliant with Islamic finance principles, might ensure that DMOs can meet varied domestic fiscal and financial goals. Not only will they foster innovation and goal achievement, but it could lower overall macro risks.

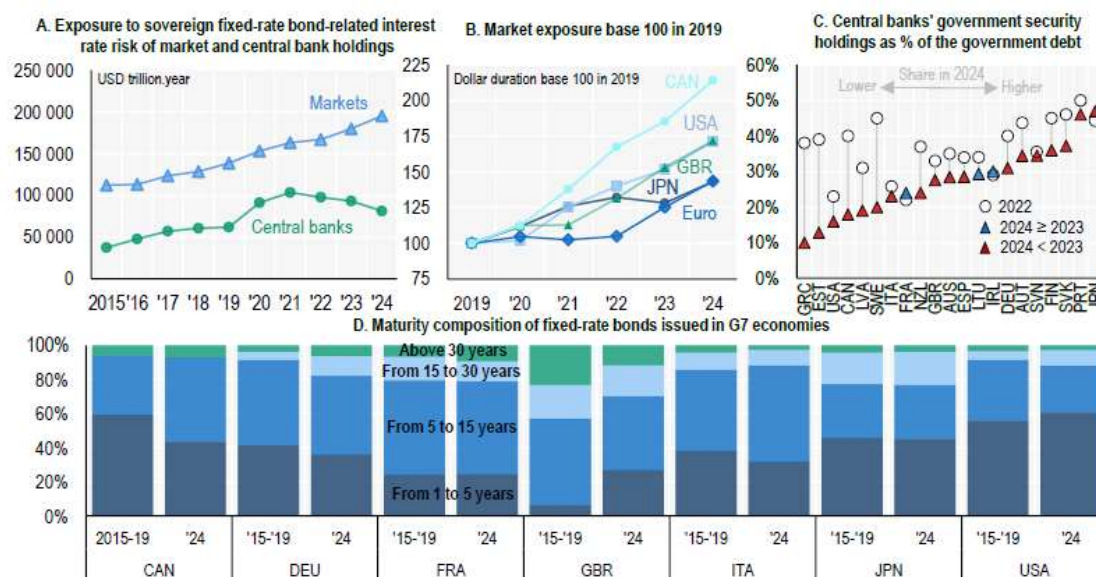
As Adam Smith notes, “He is led by an invisible hand to promote an end which was no part his intention.”¹⁹ This paper has shown how individuals maximizing their goal-specific risk-adjusted return, and likelihood of hiring skillful agents, for a given risk budget and goal, could lead provide DMOs with a simple path to overcome challenges in financing their debt.

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Appendix I – Fixed Rate Maturity Structure by Country and Central Bank Holdings

Figure 1.17. Exposure to interest rate risk, central banks' holdings and maturity composition of fixed-rate debt

Market exposure to fixed-rate bond-related interest rate risk has risen significantly



Note: Dollar duration is used to aggregate outstanding debt stocks that consist of bonds with different maturities. It measures the approximate sensitivity of a bond portfolio's value to small changes in interest rates. Specifically, it approximates how much the portfolio's value would decrease in absolute dollar terms for a one percentage point rise in interest rates. For example, a portfolio with a dollar duration of USD 10 billion would lose USD 100 million in value if interest rates increase by one percentage point (1% multiplied by USD 10 billion). The dollar duration calculations presented here are approximate: Due to the unavailability of secondary market data for all bonds, the duration of each bond was computed using the yield to maturity of the last primary market issuance. All panels cover only countries under the monetary policies of the central banks listed in Panel B, representing 93% of OECD central government outstanding marketable debt. Of these central banks, only the Bank of England has been actively selling bonds, while the others have been following a passive approach to reduce their balance sheets.

Source: 2024 OECD Survey on Central Government Marketable Debt and Borrowing; OECD (2024^[1]), OECD Economic Outlook, Volume 2024

Appendix II – Derivation of M-cube Risk Adjusted Performance

Principals will

$$\max E[r(A) - r(L)] \text{ or equation (1)} \quad (\text{A.1})$$

subject to

$$E[r(A)] = a_{\tau/L}^P \times E[r(P)] + l_{\tau/L}^L \times E[r(L)] + (1 - a_{\tau/L}^P - l_{\tau/L}^L) \times r(F) \quad (\text{A.2})$$

$$\sigma(A) = \sigma(L) \quad (\text{A.3})$$

$$\text{and TE}(A, L) = \text{TE}(\text{Target}, L) = \tau \quad (\text{A.4})$$

where $a_{\tau/L}^P$ is the allocation to the risky (agent) portfolio, P , given the goal of L and a target relative risk of T ; $l_{\tau/L}^L$ is the allocation to the goal-hedging/goal-replicating asset, L (and measures what is *de-facto* invested in the low-cost passive benchmark or “beta” in industry parlance). The super-script denotes the asset being allocated to; the subscript indicates the target relative risk (τ) and the goal-replicating asset (L). The balance of the assets is invested in the traditional absolute risk-free asset, F (and measures leverage).

Now

$$\text{TE}(A, L) = \sqrt{[\sigma^2(A) - 2 * \rho(A, L) * \sigma(A) * \sigma(L) + \sigma^2(L)]} \quad (\text{A.5})$$

Where $\rho(*)$ is the correlation parameter. From the constraint on tracking error (A.4), a unique target correlation between portfolio A and liability L , $\rho(\tau, L)$, is identified. Alternatively, instead of specifying a $\text{TE}(\text{Target}, L)$, a principal could also specify $\rho(\tau, L)$.

Substituting (A.5) in (A.4) and squaring both sides, and substituting for $\sigma(A) = \sigma(L)$ obtains

$$\rho(\tau, L) = 1 - \frac{\text{TE}(\text{Target}, L)^2}{2 \times \sigma^2(L)} = 1 - \frac{\tau^2}{2 \times \sigma^2(L)} \quad (\text{A.6})$$

The optimal allocations to “ a ” and “ l ” ensures that $\rho(A, L)$ equals $\rho(\tau, L)$ and $\sigma(A) = \sigma(L)$.

Further, calculating $\sigma^2(L) = \sigma^2(A)$ using equations (A.2) and (A.3) gives

$$\sigma^2(L) = (a_{\tau/L}^P)^2 \times \sigma^2(P) + (l_{\tau/L}^L)^2 \times \sigma^2(L) + 2 \times (a_{\tau/L}^P) \times (l_{\tau/L}^L) \times \sigma(P) \times \sigma(L) \times \rho(P, L) \quad (A.7)$$

But, the covariance of A and L , can be expressed below using the definition of covariance and equation (A.2) as

$$\sigma(A) \times \sigma(L) \times \rho(A, L) = a_{\tau/L}^P \times \sigma(P) \times \sigma(L) \times \rho(P, L) + l_{\tau/L}^L \times \sigma^2(L) \quad (A.8)$$

Using equation (A.3), and imposing $\rho(A, L) = \rho(\tau, L)$ implied from equation (A.4) as an optimal condition, equation (A.8) can be re-written as

$$\rho(\tau, L) \times \sigma^2(L) = a_{\tau/L}^P \times \sigma(P) \times \sigma(L) \times \rho(P, L) + l_{\tau/L}^L \times \sigma^2(L) \quad (A.9)$$

Dividing the RHS and LHS by $\sigma^2(L)$ and re-arranging terms,

$$l_{\tau/L}^L = \rho(\tau, L) - a_{\tau/L}^P \times \frac{\sigma(P)}{\sigma(L)} \times \rho(P, L) \quad (A.10)$$

Substituting (A.10) in (A.9) and solving gives

$$a_{\tau/L}^P = \frac{\sigma(L)}{\sigma(P)} \left[\frac{\sqrt{[1 - \rho(\tau, L)^2]}}{\sqrt{[1 - \rho(P, L)^2]}} \right] = \frac{\sigma(L)}{\sigma(P)} \left[\frac{\varphi(\tau, L)}{\varphi(P, L)} \right], \text{ where } \varphi(I, J) = \sqrt{[1 - \rho(I, J)^2]} \quad (A.11)$$

And substituting (A.10) in (A.11) gives

$$l_{\tau/L}^L = \rho(\tau, L) - \rho(P, L) \times \left[\frac{\sqrt{[1 - \rho(\tau, L)^2]}}{\sqrt{[1 - \rho(P, L)^2]}} \right] = \rho(\tau, L) - \rho(P, L) \times \left[\frac{\varphi(\tau, L)}{\varphi(P, L)} \right] \quad (A.12)$$

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⁴ <https://www.reuters.com/markets/asia/japan-issues-rare-warnings-bond-market-policy-roadmap-2025-06-13/>

⁵ https://www.dt.mef.gov.it/en/tema_del_mese/tema_mese0026.html

⁶ <https://english.dsta.nl/documents/publication/2024/12/13/outlook-2025>.

⁷ <https://www.dmo.gov.uk/about/who-we-are/>

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⁹ <https://borainvestir.b3.com.br/tipos-de-investimentos/renda-fixa/tesouro-direto/tesouro-renda-brasil-pode-ter-um-sistema-previdenciario-mais-sofisticado-que-o-de-outros-paises/>

¹⁰ <https://www.cnnbrasil.com.br/economia/tesouro-educa-novo-titulo-para-financiar-estudos-comeca-a-ser-negociado-na-b3-veja-como-funciona/>.

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¹⁵ https://deuda.mef.gub.uy/innovaportal/file/31464/1/sovereign-debt-report_june-2024.pdf

¹⁶ With a senior Treasury official asking the author why TIPS have not worked well in the US, despite all the papers issued by US Treasury to the contrary.

¹⁷ <https://www.reuters.com/article/markets/us/canada-cuts-debt-issuance-ends-sale-of-inflation-linked-bonds-idUSL1N31Z2B7/#:~:text=The%20government%20said%20it%20has,low%20demand%20for%20the%20product>.

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